

Assignment 1

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Question 1

c A1.c

```
11 #include <stdio.h>
10
9 int main() {
8     char name[100];
7     int source, destination, stations, fare;
6
5     printf("Enter passenger name: ");
4     fgets(name, sizeof(name), stdin);
3
2     printf("Enter source station number: ");
1     scanf("%d", &source);
12
1     printf("Enter destination station number: ");
2     scanf("%d", &destination);
3
4     if (destination < source) {
5         printf("Error: Destination station number cannot be less than source "
6             "station number.\n");
7         return 1;
8     }
9
10    stations = destination - source;
11    if (stations <= 5) {
12        fare = 10;
13    } else {
14        fare = 10 + (stations - 5) * 3;
15    }
16
17    printf("Passenger: %sTotal fare: ₹%d\n", name, fare);
18    return 0;
19 }
```

Output:

```
> ./A1.o
Enter passenger name: Daya Joshi
Enter source station number: 23
Enter destination station number: 45
Passenger: Daya Joshi
Total fare: ₹61
```

Question 2

```
c A1.c c A2.c
6  #include <stdio.h>
7
4  int main() {
3      int litres;
2      printf("Enter daily water consumption (litres): ");
1      scanf("%d", &litres);
7
1      printf("Usage Level   : ");
2      if (litres < 500) {
3          printf("Normal Usage\n");
4      } else if (litres ≤ 1000) {
5          printf("Moderate Usage\n");
6      } else {
7          printf("Excessive Usage\n");
8          printf("Surcharge       : Rs. 250\n");
9      }
10
11     return 0;
12 }
```

Output:

```
> gcc A2.c
> ./a.out
Enter daily water consumption (litres): 23
Usage Level   : Normal Usage
```

Question 3

c A3.c

```
15 float pA;
14 float dD;
13
12 float dC;
11
10 printf("Customer Name      :");
9  scanf("%[^\n]", name);
8
7  printf("Purchase Amount    :");
6  scanf("%f", &pA);
5
4  printf("Delivery Distance :");
3  scanf("%f", &dD);
2
1  if (dD < 0) {
21 | printf("\nInvalid Distance.");
1  }
2
3  dC = dD ≥ 5 ? 60 : 30;
4  float discAmt = pA > 3000 ? 0.1 * pA : 0;
5
6  float tP = dC + pA - discAmt;
7  float GST = tP * 0.05;
8  float fP = tP + GST;
9
10 printf("\nCustomer          : %s", name);
11 printf("\nOriginal Amount   : Rs. %.2f", pA);
12 printf("\nDiscount (10)%      : Rs. %.2f", discAmt);
13 printf("\nDelivery Charge: Rs. %.2f", dC);
14 printf("\nGST(5%)              : Rs. %.2f", GST);
15 printf("\nFinal Payable : Rs. %.2f", fP);
16
17 return 0;
18 }
```

Output:

```
> ./a.out
Customer Name      :Priya
Purchase Amount    :4000
Delivery Distance  :7

Customer          : Priya
Original Amount   : Rs. 4000.00
Discount (10)%    : Rs. 400.00
Delivery Charge: Rs. 60.00
GST(5%)           : Rs. 183.00
Final Payable : Rs. 3843.00
```

Question 4

```
c A3.c  c A4.c
33  #include <stdio.h>
32
31  int main() {
30      char name[100];
29      float cgpa, attendance;
28
27      printf("Student Name : ");
26      scanf("%s", name);
25
24      printf("CGPA : ");
23      scanf("%f", &cgpa);
22
21      printf("Attendance (%) : ");
20      scanf("%f", &attendance);
19
18      printf("\nStudent: %s\n", name);
17
16      if (cgpa > 8.5 && attendance ≥ 85) {
15          printf("Status : Eligible for Scholarship\n");
14      } else {
13          printf("Status : Not Eligible\n");
12          printf("Reason : ");
11
10          if (cgpa ≤ 8.5 && attendance < 85) {
9              printf("CGPA is below 8.5 and Attendance is below 85%%\n");
8          } else if (cgpa ≤ 8.5) {
7              printf("CGPA is below 8.5\n");
6          } else {
5              printf("Attendance is below 85%%\n");
4          }
3      }
2
1      return 0;
34 }
```

Output:

```
> gcc A4.c
> ./a.out
Student Name : Kavya
CGPA : 8.2
Attendance (%) : 90

Student: Kavya
Status : Not Eligible
Reason : CGPA is below 8.5
```

Question 5

```
c A3.c c A4.c c A5.c c A6.c
8  #include <stdio.h>
7
6  int main() {
5      int pin, correctPin = 5678;
4      int attempts = 0;
3
2      while (attempts < 3) {
1          printf("Enter PIN: ");
9          scanf("%d", &pin);
1
2          if (pin == correctPin) {
3              printf("Welcome! Your account balance is Rs. 25,000.\n");
4              return 0;
5          } else {
6              attempts++;
7
8              if (attempts < 3) {
9                  printf("Attempt %d: Incorrect PIN. %d attempt(s) remaining.\n",
10                     attempts, 3 - attempts);
11              }
12          }
13      }
14
15      printf("Card Temporarily Blocked\n");
16
17      return 0;
18  }
```

Output:

```
> gcc A5.c
> ./a.out
Enter PIN: 1111
Attempt 1: Incorrect PIN. 2 attempt(s) remaining.
Enter PIN: 2222
Attempt 2: Incorrect PIN. 1 attempt(s) remaining.
Enter PIN: 5678
Welcome! Your account balance is Rs. 25,000.
```


Question 6

```
c A3.c c A4.c c A5.c c A6.c
31 #include <stdio.h>
30
29 int main() {
28     int units;
27     float first100 = 0, next200 = 0, remaining = 0;
26     float baseBill, surcharge = 0, totalBill;
25
24     printf("Enter units consumed: ");
23     scanf("%d", &units);
22
21     if (units <= 100) {
20         | first100 = units * 2.0;
19     } else {
18         | first100 = 100 * 2.0;
17     }
16
15     if (units > 100) {
14         | if (units <= 300) {
13             | next200 = (units - 100) * 5.0;
12         } else {
11             | next200 = 200 * 5.0;
10         }
9     }
8
7     if (units > 300) {
6         | remaining = (units - 300) * 8.0;
5     }
4
3     baseBill = first100 + next200 + remaining;
2
1     if (baseBill > 2000) {
32     | surcharge = baseBill * 0.08;
1     }
2
3     totalBill = baseBill + surcharge;
4
5     printf("\nUnits Consumed : %d\n", units);
6     printf("First 100 units : Rs. %.2f\n", first100);
7     printf("Next 200 units : Rs. %.2f\n", next200);
8
9     if (units > 300) {
10     | printf("Remaining %d units : Rs. %.2f\n", units - 300, remaining);
11     }
12
13     printf("Base Bill : Rs. %.2f\n", baseBill);
14     printf("Surcharge (8%%) : Rs. %.2f\n", surcharge);
15     printf("Total Bill : Rs. %.2f\n", totalBill);
16
17     return 0;
18 }
```

Output:

```
> gcc A6.c
> ./a.out
Enter units consumed: 350

Units Consumed : 350
First 100 units : Rs. 200.00
Next 200 units : Rs. 1000.00
Remaining 50 units : Rs. 400.00
Base Bill : Rs. 1600.00
Surcharge (8%) : Rs. 0.00
Total Bill : Rs. 1600.00
```

Question 7:

```
c A3.c c A4.c c A5.c c A6.c c A7.c
25 #include <stdio.h>
24
23 int main() {
22     float temp;
21     int totalPatients = 0;
20     int feverPatients = 0;
19
18     while (1) {
17         printf("Enter temp: ");
16         scanf("%f", &temp);
15
14         if (temp == -1) {
13             break;
12         }
11
10         totalPatients++;
9
8         if (temp > 99.5) {
7             feverPatients++;
6         }
5     }
4
3     printf("\nTotal Patients Screened : %d\n", totalPatients);
2     printf("Patients with Fever : %d\n", feverPatients);
1
26     if (feverPatients > 5) {
1         printf("Warning: High number of febrile patients detected!\n");
2     } else {
3         printf("(No ward-level warning issued)\n");
4     }
5
6     return 0;
7 }
```

Output:

```
> ./a.out
Enter temp: 101.2
Enter temp: 98.4
Enter temp: 102.0
Enter temp: -1

Total Patients Screened : 3
Patients with Fever : 2
(No ward-level warning issued)
```

Question 8

c A8.c

```
8  #include <stdio.h>
7
6  int main() {
5      float distanceFloat, fuelFloat, mileageFloat;
4
3      int distanceInt, fuelInt, mileageInt;
2
1      printf("Enter distance travelled (km) : ");
9      scanf("%f", &distanceFloat);
1
2      printf("Enter fuel consumed (litres) : ");
3      scanf("%f", &fuelFloat);
4
5      mileageFloat = distanceFloat / fuelFloat;
6
7      distanceInt = (int)distanceFloat;
8      fuelInt = (int)fuelFloat;
9
10     mileageInt = distanceInt / fuelInt;
11
12     printf("\n--- Using float variables ---\n");
13     printf("Mileage : %.2f km/l\n", mileageFloat);
14
15     printf("\n--- Using int variables ---\n");
16     printf("Mileage : %d km/l (fractional part lost)\n", mileageInt);
17
18     return 0;
19 }
```

Output:

```
> ./a.out
Enter distance travelled (km) : 153
Enter fuel consumed (litres) : 9

--- Using float variables ---
Mileage : 17.00 km/l

--- Using int variables ---
Mileage : 17 km/l (fractional part lost)
```


Question 9

```
c A8.c c A9.c
14  #include <stdio.h>
13
12  int main() {
11      float temp;
10
9      while (1) {
8          printf("Enter temperature: ");
7          scanf("%f", &temp);
6
5          if (temp == 0) {
4              printf("System shutdown.\n");
3              break;
2          }
1
15         if (temp < 25) {
1             printf("%.0f°C → Fan: OFF\n", temp);
2         } else if (temp ≤ 32) {
3             printf("%.0f°C → Fan: MEDIUM speed\n", temp);
4         } else {
5             printf("%.0f°C → Fan: HIGH speed\n", temp);
6         }
7     }
8
9     return 0;
10 }
```

Output:

```
> ./a.out
Enter temperature: 22
22°C → Fan: OFF
Enter temperature: 28
28°C → Fan: MEDIUM speed
Enter temperature: 35
35°C → Fan: HIGH speed
Enter temperature: 0
System shutdown.
```

Question 10

```
c A8.c  c A9.c  c A10.c
26  #include <stdio.h>
25
24  int main() {
23      int seats = 40;
22      int choice;
21
20      while (seats > 0) {
19          printf("Book a seat? (1=Yes / 0=No): ");
18          scanf("%d", &choice);
17
16          if (choice == 0) {
15              printf("Reservation session ended.\n");
14              break;
13          }
12
11          if (choice == 1) {
10              seats--;
9              printf("Seat booked. Seats remaining: %d\n", seats);
8
7              if (seats == 0) {
6                  printf("Bus is fully booked.\n");
5              }
4          }
3      }
2
1      return 0;
27 }
```

Output:

```
) ./a.out
Book a seat? (1=Yes / 0=No): 1
Seat booked. Seats remaining: 39
Book a seat? (1=Yes / 0=No): 1
Seat booked. Seats remaining: 38
Book a seat? (1=Yes / 0=No): 0
Reservation session ended.
```